

EmacEArt Stylized Grass

URP shader for foliage cards and terrain detail meshes.

User manual — material controls and workflow.

Quick Start

The shader ships ready to paint on a Unity terrain. Three steps:

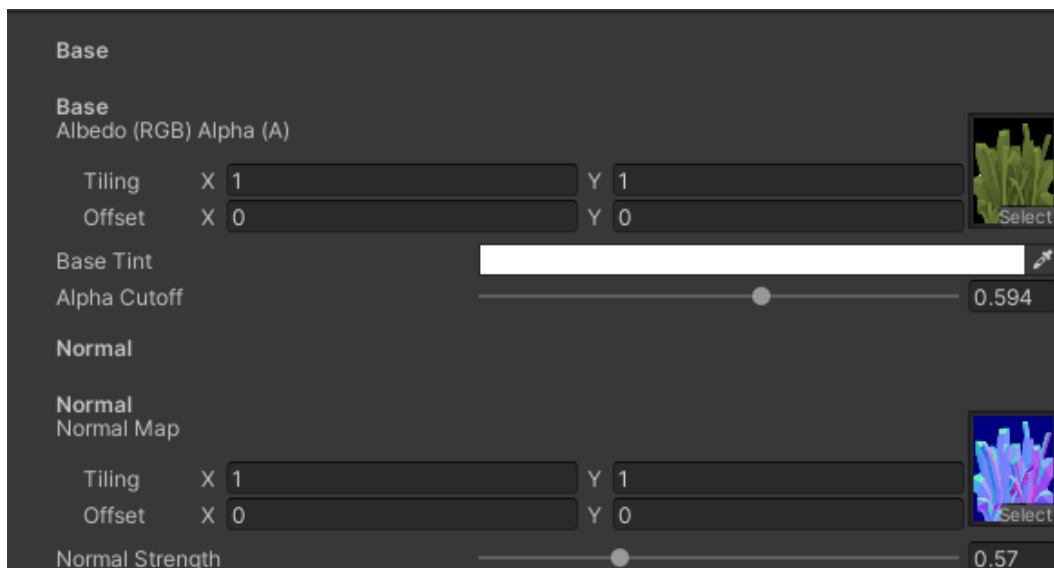
- Assign the material **EmacEArt_StylizedGrass_Mat** (or any material using the **EmacEArt/StylizedGrass** shader) to a grass card prefab.
- In the material inspector, enable **Enable GPU Instancing** at the bottom.
- On your terrain, open **Paint Details**, add the grass prefab as a **Detail Mesh**, tick **Use GPU Instancing**, then paint.

The grass will pick up wind, animated color tint and soft shadow lift automatically — out of the box, no extra textures required.

Material Inspector

The inspector is laid out top-to-bottom in the order described below. Sections without sliders shown here use Unity defaults.

Base & Normal

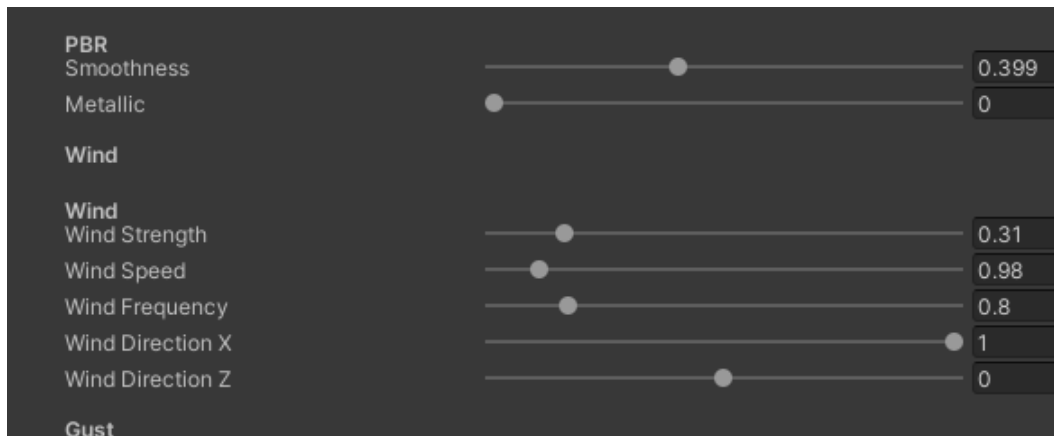


Base albedo / alpha and normal map slots.

Albedo (RGB) Alpha (A)	Grass card texture. RGB is colour, alpha is used for cutout.
Base Tint	Multiplies the albedo. Use to push the whole material warmer or cooler.
Alpha Cutoff	Threshold for the alpha test. Lower = thicker blades, higher = thinner.

Normal Map	Tangent-space normal map. Affects PBR lighting on the blades.
Normal Strength	0 flattens the normal map, 1 is the default, >1 exaggerates.

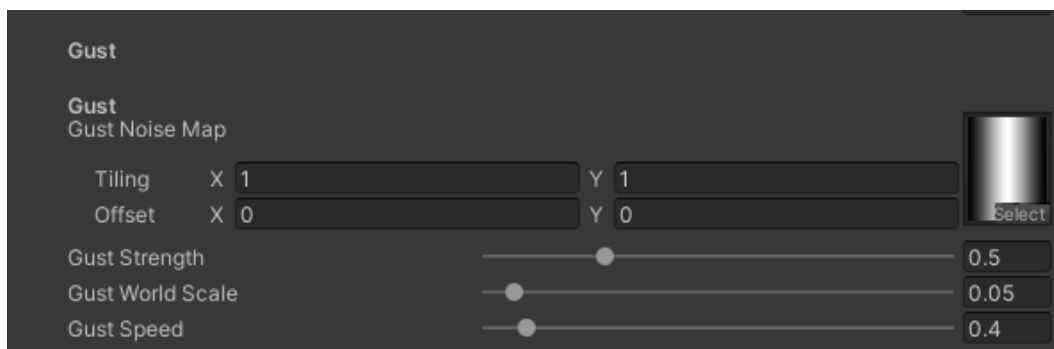
PBR & Wind



PBR surface and per-blade wind settings.

Smoothness	How glossy the blades are. Grass is usually 0.1–0.4.
Metallic	Leave at 0 unless you are doing something unusual.
Wind Strength	How far each blade can lean. 0 disables wind.
Wind Speed	Frequency of the per-blade sway in time.
Wind Frequency	Spatial frequency — how quickly the wave shifts across world space. Higher = more turbulent meadow.
Wind Direction X / Z	Horizontal wind vector. The whole meadow leans along this axis and the colour tint drifts in the same direction.

Gust



Coherent gust layer — whole regions lean together.

Per-blade wind alone looks jittery. The gust layer samples a noise texture across world XZ and scrolls it along the wind direction, so neighbouring blades receive similar values and the meadow leans in regions like a real breeze travelling across it.

Gust Noise Map	Greyscale noise texture used by the gust layer. A default noise is fine; the colour tint system below does not need this slot.
Gust Strength	How much the gust contributes on top of per-blade jitter.
Gust World Scale	Size of the gust pattern in world space. Smaller = wider gusts.
Gust Speed	How fast the gust scrolls along the wind direction.

Colour Tint

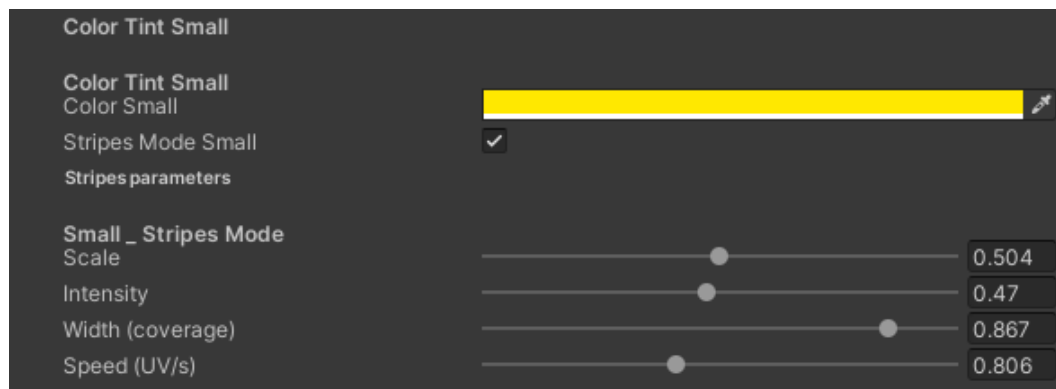
Two independent procedural layers add natural colour variation across the meadow. Both are **animated** — the pattern drifts along the wind direction set in the Wind section.

Each layer offers two modes selected by a toggle:

- **Blob mode** — organic patches drifting across the field, no dominant direction.
- **Stripes mode** — wind-driven parallel bands of varying width and spacing, perpendicular to the wind axis.

Each mode keeps its own remembered sliders. Switching the toggle does not lose tweaks you made in the other mode.

Color Tint Small



Color Tint Small

Color Tint Small
Color Small

Stripes Mode Small ☒

Stripes parameters

Small _ Stripes Mode

Scale 0.504

Intensity 0.47

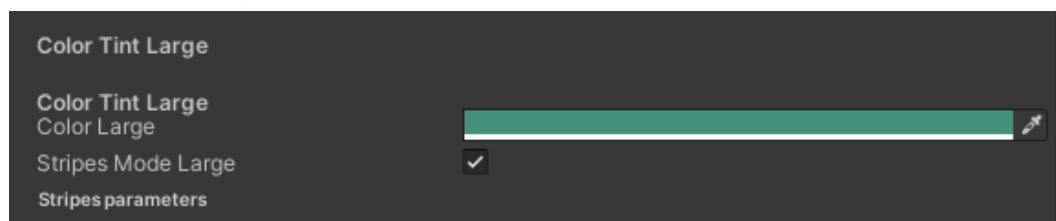
Width (coverage) 0.867

Speed (UV/s) 0.806

Small layer: fine-grained colour shifts between neighbouring tufts.

Use the Small layer for subtle blade-to-blade variation — different tufts pick up slightly different hues so the meadow doesn't read as a single flat colour. Best results with soft, warm tints close to the base colour.

Color Tint Large



Color Tint Large

Color Tint Large
Color Large

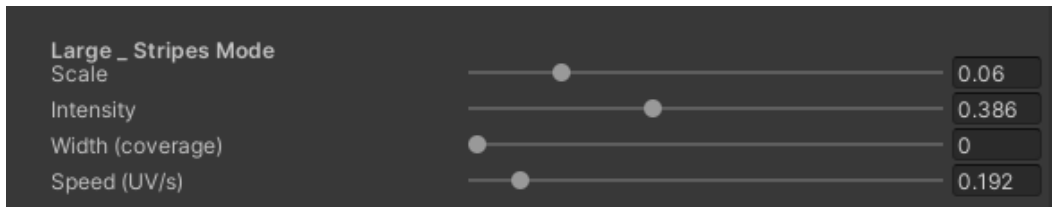
Stripes Mode Large ☒

Stripes parameters

Large layer: broad regions of colour sweeping across the field.

Use the Large layer for broad regional shifts — whole areas of the meadow drying out, distant patches catching different light. Use a muted, contrasting colour (olive, ochre, brown).

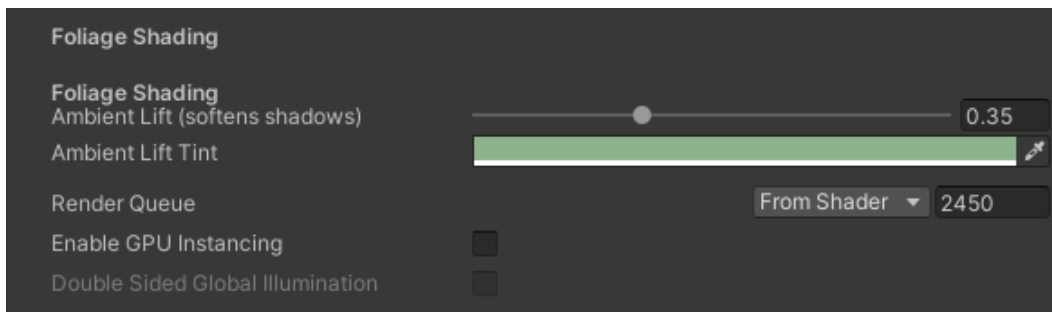
Per-layer parameters (Blob or Stripes)



The active mode's parameters are shown; the inactive mode's sliders are hidden to avoid confusion.

Color	Tint colour. White is a no-op — the layer multiplies albedo by colour, white × albedo = albedo. Pick any non-white hue to see the layer.
Stripes Mode	Toggle. Off = Blob mode (drifting patches), On = Stripes mode (parallel bands sweeping with wind).
Scale	Pattern density. Smaller value = bigger blobs / wider bands.
Intensity	Strength of the colour. 0 hides the layer, 1 saturates the peaks to the full tint colour.
Width (coverage)	How much of the surface the tint reaches. 0 = a few sharp peaks of colour, 1 = the colour covers most of the field.
Speed (UV/s)	How fast the pattern drifts along wind direction. 0 = static, 2 = fast scroll.

Foliage Shading



Shadow lift, render queue and GPU instancing toggle.

Grass cards have side-facing normals, so a top-down sun would leave them very dark in shadow. The Ambient Lift adds a subtle, albedo-tinted floor so cast shadows soften instead of crushing to black.

Ambient Lift	0 disables the lift (shadows can go very dark), 1 fully washes shadows out. 0.3–0.5 is a good starting range.
Ambient Lift Tint	Colour added in shadow. Tint slightly toward your scene's sky or GI colour for a natural look.
Render Queue	Leave at From Shader (2450, AlphaTest) unless you specifically need to push the grass earlier or later in the pipeline.

Enable GPU Instancing

Must be on for terrain Paint Details rendering. Also improves performance for prefab placement.

Painting on a Unity Terrain

The shader is designed to work with Unity's terrain Paint Details in URP. Setup:

- Select the terrain in the scene.
- Open the **Paint Details** brush in the terrain inspector.
- Click **Edit Details** → **Add Detail Mesh**.
- Drop your grass card prefab into the **Detail Prefab** slot.
- Tick **Use GPU Instancing**.
- Pick density and brush size, then paint.

If grass renders as black quads after painting, see the Troubleshooting section below.

Troubleshooting

Painted grass renders as black quads

This is the standard URP terrain detail issue. Two things must both be on:

- **Enable GPU Instancing** on the material (Foliage Shading section).
- **Use GPU Instancing** on the Detail Mesh entry in the terrain Paint Details panel.

Without both, Unity falls back to the legacy vertex-lit batch path which ignores the URP shader passes.

Tint sliders seem to do nothing

- Check the **Color** for that layer. White multiplied by anything is unchanged — pick any non-white hue and the sliders come alive.
- Verify the correct mode is active. The **Stripes Mode** toggle decides whether Blob or Stripes sliders affect the render. The other set is hidden by the custom inspector to keep the UI clean.

Grass looks too dark in shadow

Raise **Ambient Lift** in the Foliage Shading section. Adjust **Ambient Lift Tint** toward your scene's sky/GI colour so the lift reads as bounced light rather than a flat brighten.

Wind does not animate

- **Wind Strength** at 0 disables all sway. Set to 0.1–0.3 for a natural breeze.
- **Wind Speed** at 0 freezes the per-blade phase. Set to 0.5–2 for visible motion.
- Gust layer needs a noise texture in **Gust Noise Map**; without it the gust contribution is zero.

Colour tint shows a single sweeping stripe

If you are seeing a single hard front sweeping across the field, switch the layer to **Blob mode** and lower the Speed. Blob mode uses two crossed noise layers which break the visible front into organic drifting patches.